AegisCare: An Intelligent Multimodal Fall Monitoring System for Elder Care Combining Wearable Activity Recognition, Environmental Risk Mapping, PPG Vital Signs and Alert Classification

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***Abstract*—**Falls are a leading cause of injury among older adults living independently, yet most monitoring systems stop at detecting that a fall occurred, without considering the person's environment, physiological state or who should respond. This paper presents AegisCare, a low-cost, four-component Internet-of-Things (IoT) and machine-learning system for fall monitoring in elder care. (C1) A wearable BMI160 inertial node streams motion data over secured MQTT to a service that classifies six motion classes with a Random Forest and confirms a fall only when a three-gate engine (classifier confidence, trunk tilt and stay-down) agrees. (C2) ESP32 sensor nodes and an ESP32-CAM running YOLOv8n map environmental hazards (light, obstacle distance, floor wetness) into an explainable zone-level risk score. (C3) A MAX30102 fingertip photoplethysmography (PPG) sensor provides heart rate, a personalised Random Forest blood-pressure estimate and rule-based vital-sign risk status. (C4) An XGBoost classifier routes each confirmed fall to one of three responses: voice guidance for the elder, caregiver notification, or emergency escalation. Evaluated per component, C1 reached 91.97 ± 5.37% accuracy under leave-one-subject-out cross-validation on ten volunteers; C2's rule engine reproduced 18 of 18 designed test cases; C3 achieved a systolic/diastolic mean absolute error of 4.91/3.46 mmHg for one participant under similar conditions but 12.29 mmHg systolic error on an unseen activity; and C4 reached 78.47% test accuracy on a 50,000-sample synthetic dataset. The components share one MQTT/REST backbone, but an end-to-end evaluation on real falls by older adults has not yet been performed; we state the limitations of each component openly and outline the work needed before field use.

Index Terms*—fall detection, elder care, human activity recognition, IMU, environmental risk, PPG, blood pressure estimation, XGBoost, MQTT, explainable decision support.*

I. Introduction

Falls are a major public-health problem for older people. They are a leading cause of injury-related hospital admission, and the risk of falling rises with age, frailty, and exposure to hazards in the home [1], [2]. Risk factors are multifactorial: they include impaired mobility and balance, medical and cardiovascular conditions, and environmental hazards such as poor lighting, obstacles and wet floors [3], [4]. Fall detection research is extensive [5]–[8], but most deployed solutions concentrate on a single sensing modality and answer one question only: has a fall happened? They rarely combine what the body is doing, what the surroundings look like and how the person's vital signs are behaving, and they rarely decide who should respond and how urgently.

This paper describes AegisCare, a final-year research project that addresses this gap with four cooperating components built on inexpensive hardware: (C1) wearable activity recognition and fall confirmation; (C2) environmental risk mapping with a camera and IoT sensors; (C3) PPG-based heart-rate and blood-pressure monitoring; and (C4) an alert-classification engine that converts a confirmed fall and its context into a response tier. Each component was developed and evaluated by one team member; this paper describes the complete system and reports each component's results with the evaluation protocol used, so that the figures can be interpreted correctly.

The contributions are: (i) a modular architecture in which a wearable, an environmental monitor and a vital-sign sensor feed a single alert decision over a shared MQTT/REST backbone; (ii) a three-gate, explainable fall-confirmation engine evaluated on people unseen in training; (iii) a discovery-driven, additive environmental risk score with hysteresis-based actuation; (iv) a personalised PPG blood-pressure model evaluated with group-based validation against the AAMI/ISO limits; and (v) a three-tier alert classifier with a transparent analysis of its decision boundaries. We also report the limitations of each component, including the fact that the full system has not been validated end to end.

II. Related Work

Surveys classify fall detection into wearable, ambient and vision-based approaches [5]–[8]. Wearable inertial sensors are cheap and private but only describe the wearer; ambient and camera systems observe the scene but cannot measure the person's physiology; and many reported accuracies depend heavily on how data are split. Randomly splitting overlapping windows from the same recording places near-duplicate windows on both sides of the split and inflates accuracy, so subject-grouped validation is the more honest protocol [13]. Home-environment studies show that lighting, obstacles and slippery surfaces contribute to falls [3], [4], but environmental mapping is rarely integrated with wearable detection. For vision, single-stage detectors from the YOLO family make real-time person and object detection practical on commodity hardware [9], [10]. For cuffless blood-pressure estimation, PPG features can be regressed to cuff readings, but the AAMI/ISO 81060-2 standard [14] requires many participants and a strict procedure, so small studies can only offer an indication. Finally, most systems stop at binary detection; routing an event to the appropriate responder using context and a stored risk profile is under-explored. AegisCare combines these strands in one low-cost, explainable pipeline.

III. System Architecture

Fig. 1 shows the overall architecture. Three sensing components produce information about the person and the home. C1 produces FallDetected, FallType, a 0–100 RiskLevel score and a Movement grade; C2 produces zone-level environmental risk and the person's Location and Time context; C3 produces heart rate, estimated blood pressure and a Normal/Risky/Critical status. These are combined in a fused risk assessment that feeds the C4 XGBoost decision engine, which selects the notification, recovery guidance and alert type.

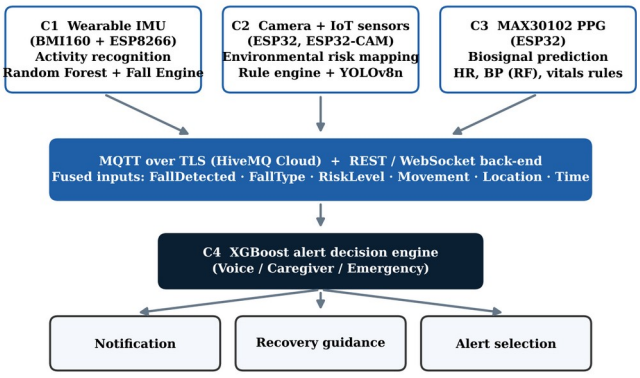


Fig. 1. AegisCare solution overview: three sensing components feed a fused risk assessment and the C4 alert decision engine.

All devices communicate over MQTT 3.1.1 [15] secured with TLS (port 8883) through a HiveMQ Cloud broker. The wearable (ESP8266) and the PPG node (ESP32) publish raw sensor data; the environmental nodes and camera announce themselves through discovery messages. Python back-end services (Flask for C1, FastAPI for C4, Python/Streamlit for C2) subscribe to the topics, run the models, and expose REST/WebSocket interfaces to React and Streamlit dashboards. The shared platform uses Supabase for stored profiles and records.

Integration status. In the current implementation the C4 classifier consumes nine features: FallDetected, FallType, RiskLevel and Movement from C1; Location and Time from C2; and Age, PreviousFalls and AssistiveDevice from a stored elder profile. C3's heart-rate and blood-pressure outputs are stored and displayed and are available to the alert service, but they are not among the nine features of the trained C4 model, and C2's zone risk score is not yet an input to C4. Fusing C2 and C3 outputs into the classifier is future work.

IV. Component Design and Methods

A. Component 1: Wearable IMU Activity Recognition and Fall Confirmation

A BMI160 six-axis IMU on a Wemos D1 Mini (ESP8266) is calibrated, smoothed with an exponential moving average, and streams at about 10 Hz over MQTT/TLS. The Flask service groups samples into 20-sample windows (about two seconds) and extracts 52 statistical features (means, standard deviations, ranges, root-mean-square, interquartile ranges and axis correlations). A Random Forest [11] classifies each window into six classes: standing, walking, and falls to the front, back, left and right. The dataset contains 14,209 samples (1,077 windows) from ten participants.

A classifier alone is not trusted to declare a fall. A stateful Fall Engine (Fig. 2) confirms a fall only when three gates agree: (1) the classifier reports a fall class with at least 60% confidence; (2) the trunk tilt computed from gravity reaches 30°; and (3) the body then remains tilted and still for two consecutive checks (about two seconds). A separate posture-decision layer grades movement on six levels (0–5), and a 0–100 fall-risk score with a template-based plain-language explanation is shown on an eight-section React dashboard.

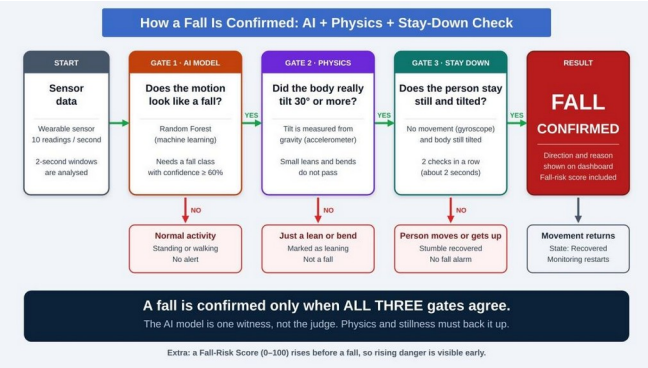


Fig. 2. C1 three-gate fall confirmation: classifier, physics (tilt) and stay-down checks must all agree.

B. Component 2: Environmental Risk Mapping

Each ESP32 sensor node carries a BH1750 light sensor, VL53L0X time-of-flight distance sensor, analogue floor-moisture sensor, PIR and MPU-6050, plus a buzzer; an ESP32-CAM streams MJPEG video. Nodes announce their identity, zone and capabilities on discovery topics, so zones and devices appear without hard-coded rooms, and a node silent for 15 s is shown OFFLINE. An additive, explainable score (maximum 100) combines light (up to 30), obstacle distance (up to 35), floor wetness (up to 30) and motion (5). The level is LOW (0–30), MEDIUM (31–60) or HIGH (61–100), and a zone takes the maximum score of its nodes. Wetness of 30% or more switches the buzzer on and 20% or less switches it off; the gap is a hysteresis band that avoids a stuttering alarm. One worker per camera runs YOLOv8n [9], [10] on the host and a Streamlit dashboard shows a dynamic 3D zone-risk map. A Random Forest risk classifier is trained offline but is not part of the live decision path.

C. Component 3: PPG Biosignal Monitoring

An ESP32 reads the raw red and infrared signals of a MAX30102 fingertip sensor at about 50 Hz and publishes them over MQTT/TLS. A Python backend filters the signal, detects heartbeats to obtain heart rate (BPM) and pulse interval, and predicts systolic and diastolic blood pressure with a personalised Random Forest trained on cuff readings from one participant (23,930 samples). Five-second averages are stored and streamed to a React dashboard over WebSockets. Because no falls or faints were recorded, fall-risk status is a rule set on one-minute averages: High (Critical) if systolic BP is above 140 or below 90 mmHg or heart rate is outside 45–120 bpm; Medium (Risky) for systolic 90–99 or 131–140 mmHg or heart rate 45–49 or 101–120 bpm; otherwise Low (Normal).

D. Component 4: XGBoost Alert Decision Engine

C4 routes every confirmed fall to one of three tiers: Voice Alert (self-guided recovery), Caregiver Alert, or Emergency Alert. A preprocessing pipeline one-hot encodes the six categorical features and passes the three numeric features through; an XGBoost classifier [12] with a softmax objective outputs class probabilities, and the highest-probability class is the decision, with its probability reported as a confidence score. The model was trained on a labelled synthetic dataset of 50,000 samples with a stratified train/validation/test split (random_state = 42) and tuned by randomized search with three-fold cross-validation, optimising macro-F1. C4 is the single decision authority: other components supply inputs, and no rule outside the classifier overrides its decision. Design revisions removed inconsistencies between independently evaluated rules and the classifier's output.

V. Evaluation Setup

Each component was evaluated on its own data and with the protocol appropriate to it (Table I). C1 used three protocols: a random 80/20 window split (P1, optimistic because of window overlap), leave-one-subject-out cross-validation (LOSO, P2, the main result), and an 80/20 split by person with two participants held out (P3). C3 used group-based validation: five-fold time-block and leave-one-activity-segment-out, compared with a mean-predicting baseline and with the AAMI limits (mean error within ±5 mmHg, SD within 8 mmHg) [14]. C2 used an 18-case offline rule test and a stratified 80/20 split on a synthetic dataset for its Random Forest baseline. C4 used a held-out stratified test set (n = 6,039) never used in tuning [13].

TABLE I. Evaluation Basis of Each Component

Comp.	Data	Protocol
C1	14,209 samples, 1,077 windows, 10 volunteers (simulated falls)	LOSO CV; person-split; random split
C2	24,000 synthetic samples; 21 real samples (1 node)	18 rule tests; stratified 80/20
C3	23,930 samples, 1 participant, cuff reference	Time-block and unseen-activity validation
C4	50,000 synthetic samples	Held-out stratified test, n = 6,039

VI. Results

A. C1: Activity Recognition and Fall Confirmation

Table II shows that a random window split overstates accuracy: the Random Forest scores 93.52% on a random split but 91.97 ± 5.37% on people it has never seen. Accuracy varies strongly between people, from 81.65% (S03) to 99.39% (S01). The Random Forest and KNN are practically tied (0.26-point difference, less than one standard deviation) and both outperform the Decision Tree. The Random Forest was deployed for its highest mean LOSO

accuracy among the classical models. Pooling out-of-fold predictions gives 92.39% accuracy (weighted F1 92.37%, macro F1 90.8%); the four fall classes have F1 between 85.1% and 94.3%, with forward falls weakest (F1 85.1%) and standing still the lowest-recall class (86.0%).

TABLE II. C1 ACCURACY (%) BY EVALUATION PROTOCOL

Model	Random split	LOSO (mean ± SD)	Person split
Random Forest	93.52	91.97 ± 5.37	92.12
Decision Tree	89.35	83.57 ± 8.04	86.70
KNN	94.44	91.71 ± 4.63	91.13

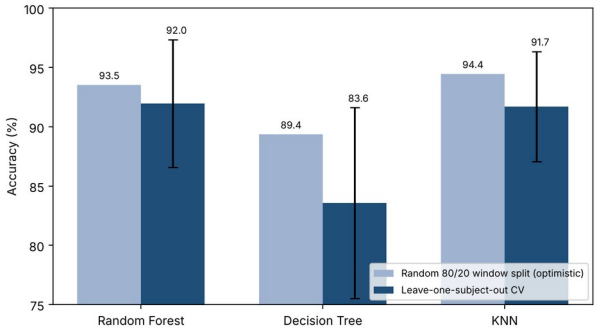


Fig. 3. C1 accuracy under a random window split (light) versus leave-one-subject-out validation (dark, bars show ±1 SD).

Most fall recordings were shorter than one window and had to be zero-padded; accuracy is 92.96% on the 767 unpadded windows and 90.97% on the 310 padded windows. Of 102 planned test cases, 25 were executed (23 passed, 2 failed); the failures revealed a confirmation weakness and an API error-handling defect, and the remaining cases are pending.

B. C2: Environmental Risk Mapping

All 18 offline rule tests passed; a node with 15 lux (+30), 450 mm (+20), 60% wetness (+20) and motion (+5) scores 75 (HIGH) and the maximum of 100 is reproduced. The offline Random Forest baseline, trained on 24,000 synthetic samples, reached 79.4% accuracy on 4,800 test samples (macro F1 0.798), with no LOW sample predicted as HIGH and no HIGH as LOW; MEDIUM was the hardest class (recall 0.654) (Table III). Because the synthetic labels disagree with the rule engine in about half of cases and the moisture direction is opposite to the real sensor, this figure measures how well the model learned the synthetic generator, not real environmental risk. Verification also exposed a missing-distance handling defect with a one-line fix.

TABLE III. C2 AND C4 PER-CLASS TEST RESULTS

Class	Prec.	Recall	F1
C2 LOW	0.782	0.896	0.835
C2 MEDIUM	0.795	0.654	0.717
C2 HIGH	0.806	0.880	0.841
C4 Voice Alert	0.79	0.88	0.83
C4 Caregiver Alert	0.71	0.66	0.68
C4 Emergency Alert	0.87	0.82	0.84

C. C3: PPG Vital Signs

Table IV shows that, for one participant under conditions similar to training (time-block validation), the Random Forest clearly outperforms a mean-predicting baseline (systolic MAE 4.91 vs. 11.65 mmHg; diastolic 3.46 vs. 7.25 mmHg) and lies inside the AAMI limits (systolic mean error 0.67, SD 6.37 mmHg; diastolic 0.38, SD 4.48 mmHg). On an activity not seen in training the systolic MAE rises to 12.29 mmHg (R^2 = 0.026) and falls outside the limits (mean error 6.13, SD 12.65 mmHg). The model depends mainly on average heart rate (79% importance), and the raw PPG values add little. This comparison is only a guide: the AAMI/ISO protocol needs many participants, and this study used one.

TABLE IV. C3 BLOOD-PRESSURE MAE (MMHG), ONE PARTICIPANT

Validation	Model	Systolic	Diastolic
Time-block	Baseline	11.65	7.25
Time-block	Random Forest	4.91	3.46
Unseen activity	Baseline	16.82	10.14
Unseen activity	Random Forest	12.29	7.34

D. C4: Alert Classification

The tuned XGBoost model reached 78.47% test accuracy (baseline 78.34%) and narrowed the train–test gap from 2.72 to 1.06 percentage points; macro precision, recall and F1 were 78.82%, 78.69% and 78.60%. Emergency Alert had the highest precision (0.87), Voice Alert the highest recall (0.88), and Caregiver Alert the lowest recall (0.66) because it lies between the other two classes (Table III). A class-weighted variant lowered F1 for Caregiver Alert and was excluded. Feature importance is dominated by C1 outputs: RiskLevel 38.25%, FallDetected 25.30% and FallType 21.10%, with Location 5.12% and Time 3.78% from C2 and the elder profile features each below 1.4%. On confirmed falls in the data, 67.3% were routed to Emergency, 31.1% to Caregiver and 1.6% to Voice Alert. Because the dataset is synthetic, these figures describe the labelling scheme and not real-world response needs.

VII. Discussion and Limitations

Three design choices recur across the components. First, decisions are made explainable: the three gates, the additive environmental score and the vital-sign rules can each be inspected, and C4 exposes a confidence score. Second, a model is never the sole judge: C1 requires physical and temporal confirmation, and C4 is the single decision authority. Third, results are reported with the protocol that produced them, since window-overlap leakage and synthetic data can make accuracy look better than it is.

The limitations are substantial and should be read before any claim of effectiveness. (1) No end-to-end evaluation: no real falls by older adults have been recorded, the combined pipeline has not been tested as a whole, and no user study was performed. (2) C1 used ten healthy volunteers with simulated falls; most fall recordings required zero-padding; the 30° tilt gate may miss shallow falls; 77 of 102 test cases remain unrun. (3) C2's classifier was trained on synthetic data, thresholds are engineering choices, tests used one node, one camera and one zone, and certificate verification is disabled in the development setup. (4) C3 used one participant; its blood-pressure model fails on an unseen activity, and fall-risk rules are untested on real falls or faints. Message delivery from the PPG node was 84.67% in one system test, so loss must be handled before clinical use. (5) C4 was trained and evaluated on synthetic data whose labels come from the same assumptions as the features, so 78.47% accuracy does not predict real-world performance, and C2 and C3 outputs are not yet among its inputs. (6) The blood-pressure estimate is not a medical-grade measurement and the system is not a certified medical device.

VIII. Conclusion and Future Work

AegisCare shows that a low-cost, explainable fall-monitoring pipeline can combine wearable activity recognition, environmental risk mapping, PPG vital signs and context-aware alert classification on a shared MQTT/REST backbone. On unseen people, the wearable classifier reached about 92% accuracy; the environmental rule engine reproduced all designed test cases; the personalised PPG model was accurate for one person under similar conditions but not on an unseen activity; and the alert classifier reached 78.47% on synthetic data. These are component-level proof-of-concept results.

Future work includes: collecting real falls and daily activity from older adults under ethical approval; running an end-to-end evaluation; replacing synthetic C4 labels with clinician-reviewed or real-world outcomes and adding C2 zone risk and C3 vital-sign features to the classifier; using full-length fall windows to remove zero-padding; training a custom hazard detector for C2; validating C3 on multiple participants and activities under the full AAMI/ISO protocol; and securing the deployment with certificate validation, per-device credentials and authenticated dashboards.

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